Orbital-Optimized DFT for Excited States in MOUSE:ΔSCF, Spin-Pure ROKS, and Analytic Nuclear Response

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Draft companion manuscript — 2 November 2026

Abstract

We present a full-integral DFT implementation of state-specific excited-state optimization in MOUSE. Fixed-occupation unrestricted ΔSCF and a common-orbital two-open-orbital ROKS construction are implemented on top of the ground-state PySCF machinery. The ROKS energy is the multiplet combination $E_S = 2E_M - E_T$ evaluated with common spatial orbitals, giving a coherent spin-pure singlet representation. Orbital stationarity is solved with a residual trust-region algorithm and its coupled response is differentiated analytically through molecular gradients and Hessians, including moving-grid XC terms. The present draft establishes the implementation and derivative qualification; chemical accuracy, broad transferability, and a complete comparison with ORCA MOM/GMF and native PySCF MOM remain future work.

1 Scope and motivation

Response methods such as TDA and TD-DFT describe many excited states from a ground-state reference. State-specific orbital optimization instead obtains an excitation from the difference between stationary electronic energies and can reuse the energy and derivative infrastructure of a ground-state program. Levi's direct-optimization and saddle-search work motivates treating an excited state as a stationary point, often a saddle, rather than as a fragile SCF fixed point. The present work develops that idea as a general full-DFT backend, separate from the semiempirical OO-CID model reported elsewhere.

2 Theory

For a fixed non-Aufbau occupation, unrestricted ΔSCF optimizes the spin orbitals directly and may remain spin contaminated. For a two-open- orbital singlet, MOUSE uses common spatial orbitals and evaluates

$$E_S[\mathbf{C}] = 2E_M[\mathbf{C}] - E_T[\mathbf{C}], \tag{1}$$

where mixed and triplet components share the same orbitals. The resulting two-determinant CSF has $\langle S^2 \rangle = 0$ by construction, while the DFT multiplet-sum energy remains an approximate functional expression.

3 Implementation

The implementation is a PySCF extension with fixed occupations, state following, orbital continuation, and optional freeze-and-release preparation. The final solve releases all nonredundant orbital rotations. The residual Jacobian is treated as nonsymmetric away from stationarity; its exact adjoint drives a small residual trust-region step. Range-separated exact integrals are cached when memory permits, with an exact native fallback.

3.1 Full-DFT implementation of the orbital response

A separate reference implementation reuses molecular UHF/UKS energies, Fock matrices and analytic density-response kernels from PySCF.Sun et al. [2020] Fixed-occupation unrestricted ΔSCF and common-orbital ROKS solve orbital stationarity with Newton–Krylov steps, retaining negative curvature without a TDA calculation or MOM reassignment. ROKS uses a residual trust region and retained-target recovery; final stationarity and response release every nonredundant rotation (SI). The coherent ROKS CSF is a pure singlet, while its common-orbital $2E_M - E_T$ functional energy remains approximate. Neither an energy offset nor semiempirical kernel parameters enter this implementation.

The analytic nuclear Hessian includes the complete noncanonical energy-weighted density, coupled common-orbital response, moving Becke-grid derivatives, AO derivatives through third order and LibXC vxc/fxc. Water and formaldehyde checks cover HF, LDA, PBE0 and CAM-B3LYP; DFT Hessian–vector errors are below 5.1×10^{-9} Ha/bohr^2. Cached grid intermediates and the revised solver preserve complete Hessians within 3.1×10^{-13} Ha/bohr^2. The SI documents retained-target recovery and derivative certificates. These small-basis implementation checks establish neither spectral accuracy nor a speed advantage over MOM, and do not extend the semiempirical accuracy ledger.

4 Qualification and current limitations

The qualified scope is all-electron, full-integral molecular HF and LDA/GGA, including global and range-separated hybrids, with analytic moving-grid response. Density fitting, solvent, relativistic, meta-GGA and ECP Hessians are outside the present qualification. The tests establish derivative consistency and solver robustness; they do not yet establish a general accuracy advantage over MOM, ORCA, PySCF, TDA, or other response methods.

5 Outlook

The next stage will compare identical roots and functionals across MOUSE, ORCA and native PySCF, add GMF/saddle-following benchmarks, and evaluate excited-state geometries, frequencies, CT and Rydberg states on a frozen independent set.

References

Qiming Sun, Xing Zhang, Samragni Banerjee, et al. Recent developments in the PySCF program package. *The Journal of Chemical Physics*, 153:024109, 2020. doi: 10.1063/5.0006074.